5to. Congreso Internacional de Tecnología y Ciencias Aplicadas (CITCA)

T e c N M / C E N I D E T

Fecha: 28 de noviembre 2025

Hora	Actividad	Lugar
08:00 - 14:00	**Registro de asistentes**	Explanada del Auditorio
09:00 - 10:00	**Ponencia Magistral 1: Celdas de combustible PEM y AEM: De la ciencia a la aplicación.** **Dra. Tatiana Romero Castañón.** Moderador: ----------	Auditorio
10:00 - 12:00	**Presentación de pósteres**	Explanada del Auditorio

Ponencias Sala 1, Sesión 1: Eficiencia Energética y Energías Renovables Moderador: DIE					
10:00 - 10:20	Virtual	CITCA2025-02-0312	Simulation of the chlorobenzene production process using DWSIM	Maritza Jaquelline Huazo Mejia	Sala 1: Auditorio
10:20 - 10:40	Presencial	CITCA2025-02-0067	Peltier cell for generating energy using passive sources of cold and heat	Oswaldo Aparicio Rodriguez	
10:40 - 11:00	Presencial	CITCA2025-02-0319	Study and evaluation of a reaction chamber for heat treatments of semiconductor materials by electromagnetic induction.	Jorge Edmundo Mastache Mastache	
11:00 -11:20	Presencial	CITCA2025-02-0330	Solar Irradiance Estimation using Extended Kalman Filter: Application in Chicoloapan, Mexico	Michael Jossue Rojas García	
11:20-11:40	Virtual	CITCA2025-02-0303	Sustainable assessment of extractive distillation for ethanol dehydration using ionic liquids as entrainers	Lizeth Sarai Ávila Loaiza	
11:40 - 12:00	**RECESO**				
12:00 - 13:00	**Ponencia Magistral 2: Modelos neuronales de tercera generación y su aplicación en el reconocimiento de patrones** **Dr. Roberto Antonio Vázquez Espinoza.** **Moderador: ----------**				**Auditorio**
Ponencias Sala 1, Sesión 2: Educación y Cultura Moderador: DIE					
13:00 -13:20	Virtual	CITCA2025-02-0070	Impact of Interconnect Crossings on the Performance of Rotary Traveling-Wave Oscillator Arrays	Rogelio Manuel Higuera González	Sala 1: Auditorio
13:20 - 13:40	Presencial	CITCA2025-02-0133	Frugal Prototype of an Autonomous Robot with Sensory Fusion and Bluetooth Communication	Ricardo Álvarez González	
13:40 -14:00	Virtual	CITCA2025-02-0346	Tecnogol: Fostering Ethics, Teamwork, and Love for Engineering through Competitive Robotics	Mónica Graciela Marín Amaya	
14:00 - 14:20	**RECESO**				
14:20 - 14:40	Virtual	CITCA2025-02-0236	Mechanical Behavior Analysis of a Composite Material Under Tensile Load: Simulation vs. Experimental Test	Lazarin Leaño José Manuel	
14:40 - 15:00	Virtual	CITCA2025-02-0250	Flight Envelope Calculation for Low-Speed Aircraft	Ian Yaocelotl Alvarado Jiménez	
15:00 - 15:40	Virtual	CITCA2025-02-0194	AquaGuard: An Interactive Website on the Quality of Surface Water Bodies in Mexico	Esteban Valente Arenas	

Ponencias Sala 2, Sesión 1: Sistemas agroalimentarios sustentables Moderador: CCI					
10:00 - 10:20	Virtual	CITCA2025-02-0061	Evaluation of the antifungal effect of a zinc nanoparticle formulation (HMC) against Fusarium solani during germination and plant development of tomato (Solanum lycopersicum L.)	Dalia Elizabeth Miranda Castilleja	Sala 2: Aula 2102 Edificio del Departamento de Ingeniería Electrónica
10:20 - 10:40	Virtual	CITCA2025-02-0130	Improvements in Asexual Propagation of Stevia rebaudiana Bertoni Through the Use of Biostimulants in Michoacan, Mexico	Juan Carlos Álvarez Hernández	
10:40 - 11:00	Virtual	CITCA2025-02-0173	Postharvest Efficiency and Cup Quality of Marsellesa Coffee Under Traditional Processing in Sierra Mixe, Oaxaca	José Uriel García Domínguez	
11:00 -11:20	Virtual	CITCA2025-02-0220	Design and Validation of a High-Resolution Image Capture System for Focus Stacking of Agricultural Plant Leaf Images	Diana Karina Jacobo Rubio	
11:20 - 11:40	Virtual	CITCA2025-02-0280	"AVISENS – Development of a Real-Time Monitoring System for Environmental Variables in Laying Hens"	Montserrat Aimeé Santos Bravo	
11:40 - 12:00	**RECESO**				

12:00 - 13:00	**Ponencia Magistral 2: Modelos neuronales de tercera generación y su aplicación en el reconocimiento de patrones Dr. Roberto Antonio Vázquez Espinoza.**				**Auditorio**
Ponencias Sala 2, Sesión 2: Sistemas agroalimentarios sustentables Moderador: CCI					
13:00 -13:20	Virtual	CITCA2025-02-0059	Earthtics, Prototype Design with NPK Sensor for Measuring Nutrients and Soil Parameters.	Andrea Guadalupe Vela Peraza	Sala 2: Aula 2102 Edificio del Departamento de Ingeniería Electrónica
13:20 - 13:40	Virtual	CITCA2025-02-0102	Evaluation of Tomato (Solanum lycopersicum L.) Hydroponic Effluents as a Growth Medium for the Propagation of Azolla (Salviniaceae)	Saul Santiago Bravo	
13:40 -14:00	Virtual	CITCA2025-02-0117	Early morphometric response of potato inoculated with Rhizobium sp. ACO-34A under saline stress: a sustainable biofertilization approach	Erick Fabián Sarmiento Megchum	
14:00 - 14:20	**RECESO**				
14:20 - 14:40	Virtual	CITCA2025-02-0118	Initial growth of carrot (Daucus carota L.) in response to biofertilization with Rhizobium	Stephanie Solís Zebadúa	
14:40 - 15:00	Presencial	CITCA2025-02-0019	Environmental sustainability-agronomic indicators of indoor crops in urban agriculture	Jorge Flores Velázquez	
15:00 - 15:40	Virtual	CITCA2025-02-0135	From Productivity to Sustainability: Coffee Cluster Development in the Sierra Negra of Puebla	Faustino Sergio Villafuerte Palavicini	

Ponencias Sala 3, Sesión 1: Salud Moderador: DCC					
10:00 - 10:20	Virtual	CITCA2025-02-0238	Fuzzy Inference Modeling for Estimation of Boarding and Alighting Times in Passengers with Limited Mobility	Ricardo Pérez Cabrera	Sala 3: Aula 2109 Edificio del Departamento de Ingeniería Electrónica
10:20 - 10:40	Virtual	CITCA2025-02-0264	Compy: Intelligent Wearable System for Assisted Navigation in Visually Impaired Individuals	Brayan De Jesús Lozano Rendón	
10:40 - 11:00	Presencial	CITCA2025-02-0147	Prototype and Pilot Evaluation of a Non-Invasive Multicolor Optical Method for Postprandial Glucose Detection in Dried Saliva Using KNN Classification	Miguel Sánchez Brito	
11:00 -11:20	Virtual	CITCA2025-02-0342	Intelligent Assistive System for the Early Detection and Mitigation of Hallux Valgus and Flat Foot Using Computer Vision	Nancy Daniela García Morales	
11:20 - 11:40	Virtual	CITCA2025-02-0315	Pressure Ulcer Prevention With 3D-Printed in Lattice Structure with Biocompatible Propieties Derived from Tio? Nanoparticles	Victor Miguel Saucedo Bocanegra	
11:40 - 12:00	**RECESO**				
12:00 - 13:00	**Ponencia Magistral 2: Modelos neuronales de tercera generación y su aplicación en el reconocimiento de patrones Dr. Roberto Antonio Vázquez Espinoza. Moderador: ----------**				**Auditorio**
Ponencias Sala 3, Sesión 2: Salud Moderador: ----					
13:00 -13:20	Virtual	CITCA2025-02-0337	Development of a Multifunctional Wheelchair to Support Patient Transfer and Rehabilitation	Gerardo Jiménez González	Sala 3: Aula 2109 Edificio del Departamento de Ingeniería Electrónica
13:20 - 13:40	Virtual	CITCA2025-02-0334	Software Development for External Breast Prostheses	Mario Adán Pérez Frías	
13:40 -14:00	Presencial	CITCA2025-02-0215	Security System for Medical Imaging Based on Digital Steganography and Biometrics	Reyes Gordillo Jonathan Daniel	
14:00 - 14:20	**RECESO**				
14:20 - 14:40		CITCA2025-02-0356	Active monitoring with honey for diabetic foot regeneration		
14:40 - 15:00		CITCA2025-02-0011	Design and Virtual Validation of a Continuous Passive Motion Mechanism for Ankle Rehabilitation Based on Lissajous Curves		
15:00 - 15:40	Presencial	CITCA2025-02-0184	Electronic System for treating sarcopenia	Victor Hugo Ortiz	

Ponencias Sala 4, Sesión 1: Tecnología 5.0 y Ciencia de Datos Moderador: DCC					
10:00 - 10:20	Presencial	CITCA2025-02-0023	Literature Review: Detection and Estimation of Foliar Severity in Tomato (Solanum lycopersicum)	Astrid Ariadna Torres Fernández	Sala 4: Aula 2111 Edificio del Departamento de Ingeniería Electrónica
10:20 - 10:40	Virtual	CITCA2025-02-0008	Study and Coverage Analysis of a LoRa Transmission System at UNAM's University City for IoT Applications.	Roberto Travieso Valdés	
10:40 - 11:00	Virtual	CITCA2025-02-0030	Graph Data Analysis Against Money Laundering	Alfredo García Castañón	
11:00 -11:20	Presencial	CITCA2025-02-0287	Dynamic Drone Mesh Networks for Scalable Quantum Key Distribution in Challenging Environments	Hernández Barreto Alan Daleth	
11:20 - 11:40	Virtual	CITCA2025-02-0306	Platform Intercomparison for VWC Estimation:Assessing NDVI Profiles from Sentinel-2 and MODIS for Maize Monitoring in Tlaxcala.	María Fernanda Soria Rodríguez	

11:40 - 12:00	**RECESO**				
12:00 - 13:00	**Ponencia Magistral 2: Modelos neuronales de tercera generación y su aplicación en el reconocimiento de patrones** **Dr. Roberto Antonio Vázquez Espinoza.**				**Auditorio**
Ponencias Sala 4, Sesión 2: Tecnología 5.0 y Ciencia de Datos Moderador: CCIMK					
13:00 -13:20	Presencial	CITCA2025-02-0124	Clapp oscillator implementing Knowm memristors	Tomás Alberto Jiménez Castillo	Sala 4: Aula 2111 Edificio del Departamento de Ingeniería Electrónica
13:20 - 13:40	Virtual	CITCA2025-02-0212	Mechanical Design of an Ergonomic Cabin for a Production Line Study	Alejandra Cruz Cruz	
13:40 -14:00	Virtual	CITCA2025-02-0231	Kinematic modeling, simulation, and validation of a 4-DOF robotic manipulator in ROS2	Enrique García Trinidad	
14:00 - 14:20	**RECESO**				
14:20 - 14:40	Virtual	CITCA2025-02-0081	Design and Implementation of a Balanced Scorecard in a Human Capital Services Company: the Be Group Case Study	Cesar Antonio Campos Villegas	
14:40 - 15:00	Virtual	CITCA2025-02-0128	TDS Sensor Analysis with Artificial Neural Network for Water Quality Evaluation	Gilbert Francis Pérez García	
15:00 - 15:40		CITCA2025-02-0335	Evaluation of the acceptance of a new snack-type product (churros) made from pulque agave flour (Agave salmiana)		

Ponencias Sala 5, Sesión 1: Inteligencia Artificial Moderador: DCC					
10:00 - 10:20	Virtual	CITCA2025-02-0031	Comparative Study Of Natural Language Processing Algorithms For Topic Analysis In Identifying Issues Faced By Higher Education Students Based On Unstructured Texts	Xavier De Jesús López Velasco	Sala 5: Aula 3101 Edificio del Departamento de Ciencias Computacionales
10:20 - 10:40	Presencial	CITCA2025-02-0028	Medical-AId, a platform for medical services assisted by mobile, web, and AI devices	David Arnoldo Valtiérrez Angel	
10:40 - 11:00	Virtual	CITCA2025-02-0172	Autonomous Navigation System with Computer Vision and Fuzzy Logic for Mobile Robots using ESP32-CAM	Rosa Adriana Jiménez López	
11:00 -11:20	Presencial	CITCA2025-02-0199	Data Augmentation Techniques for Images Containing People: Review of Methods, Efficiency, and Practical Challenges	Pedro Arcos Méndez	
11:20 - 11:40	Virtual	CITCA2025-02-0214	Artificial vision system for drowsiness and distraction detection in high-risk activities	Sheila Mishel Márquez Marina	
11:40 - 12:00	**RECESO**				
12:00 - 13:00	**Ponencia Magistral 2: Modelos neuronales de tercera generación y su aplicación en el reconocimiento de patrones** **Dr. Roberto Antonio Vázquez Espinoza.**				**Auditorio**
Ponencias Sala 5, Sesión 2: Inteligencia Artificial Moderador: DCC					
13:00 -13:20	Presencial	CITCA2025-02-0146	Use of artificial intelligence in adaptation of gamification in scholar candidate selection	Mónica Cristal Girón Colín	Sala 5: Aula 3101 Edificio del Departamento de Ciencias Computacionales
13:20 - 13:40	Virtual	CITCA2025-02-0226	Genetic Algorithm to Maximize the Strength of Microalloyed Steel	Jazmín Yanel Juárez Chávez	
13:40 -14:00	Presencial	CITCA2025-02-0295	Automatic Design of a Simple Functional Neural Network	Antonio Aguilar Patricio	
14:00 - 14:20	**RECESO**				
14:20 - 14:40	Virtual	CITCA2025-02-0033	Anomaly Detection in Heart Sounds Using Daubechies Wavelet Analysis and MFCCs Through Machine Learning Techniques	Sebastian Guzman Alfaro	
14:40 - 15:00	Presencial	CITCA2025-02-0222	Challenges in the Construction of a Dataset for Urban Wind Speed Forecasting	William Flores Gomez	
15:00 - 15:40	Virtual	CITCA2025-02-0062	Evaluation of Fuzzy Systems for Soil pH Prediction in Papaya Crops	Carlos David Echevarría Lezcano	

Ponencias Sala 6, Sesión 1: Desarrollo Sostenible Moderador: CCIMK					
10:00 - 10:20	Virtual	CITCA2025-02-0039	Implementation of an Automated Integrated Waste Extraction and Lubrication System for CNC Machines.	José Yahir Rosas García	Sala 6: Aula 3102 Edificio del Departamento de Ciencias Computacionales
10:20 - 10:40	Virtual	CITCA2025-02-0074	Detection of the Varroa destructor mite through computer vision in Apis mellifera	Daniel Mauricio Matías García	
10:40 - 11:00	Presencial	CITCA2025-02-0213	Earthworm Humus: Biology, Technology, and Opportunities for a Sustainable Fertilizer	Tlaloc González Sánchez	
11:00 -11:20	Virtual	CITCA2025-02-0075	A preliminary system for sound and environmental monitoring in beehives using LoRa transmission.	Juan Antonio Villa Guerrero	
11:20 - 11:40	Virtual	CITCA2025-02-0016	Design of an Automated Intelligent Irrigation System for a Persian Lime Agricultural Field.	Yutzil Osorio Sánchez	
11:40 -12:00	**RECESO**				

12:00 - 13:00	**Ponencia Magistral 2: Modelos neuronales de tercera generación y su aplicación en el reconocimiento de patrones** **Dr. Roberto Antonio Vázquez Espinoza.**				**Auditorio**
Ponencias Sala 6, Sesión 2: Desarrollo Sostenible y Gestión del Agua Moderador: CCI					
13:00 -13:20	Virtual	CITCA2025-02-0327	Thermal compensation of a Multimode-Interference-based fiber optic sensor for water quality measurements	Víctor Iván Ruiz Pérez	Sala 6: Aula 3102 Edificio del Departamento de Ciencias Computacionales
13:20 - 13:40	Presencial	CITCA2025-02-0225	Characterization of vernacular materials: case of Adobe	Celeste Bello Pastor	
13:40 -14:00		CITCA2025-02-0349	Study of Expired Medications as Copper Corrosion Inhibitors.		
14:00 - 14:20	**RECESO**				
14:20 - 14:40		CITCA2025-02-0270	Hand Prototype For Catheterization Practice Of Nursing Students		
14:40 - 15:00					
15:00 - 15:40					

16:00 **Clausura y entrega de reconocimientos ganadores**